



B.M.S. COLLEGE OF ENGINEERING Bengaluru-560019.

Autonomous College, affiliated to
Visvesvaraya Technological University, Belagavi



Project Work-2 Report on
**“A Wearable EMG-Morse Code Silent Communication with AES
Security”**

Submitted in partial fulfilment of the requirement for completion of
PROJECT WORK-2 [23EC7PWPJ2]

Submitted by

RITHIKA P M	1BM22EC198
 AISIRI G AITHAL	1BM23EC400
 DHANANJAY L	1BM23EC404
 M ADARSHA	1BM23EC410

Under the guidance of

Dr. Lalitha S

Associate Professor

BMS College of Engineering

Bengaluru-560019

Academic Year

2025 – 2026

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BMS College of Engineering
Bull Temple Road, Basavanagudi, Bengaluru-560019

BMS COLLEGE OF ENGINEERING

Autonomous college, affiliated to VTU

Bull Temple Road, Bengaluru – 560 019

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



CERTIFICATE

This is to certified that the Project work-2 entitled “**A Wearable EMG-Morse Code Silent Communication with AES Security**” is a bonafide work carried out by **Rithika P M (1BM22EC198), Aisiri G Aithal (1BM23EC400), Dhananjay L (1BM23EC404), M Adarsha (1BM23EC410)**, submitted in partial fulfilment of the requirement for completion of PROJECT WORK-2 [23EC7PWPJ2] of Bachelor of Engineering in Electronics and Communication during the academic year 2025-26. The Project Work-2 report has been approved as it satisfies the academic requirements.

Guide

Dr. Lalitha S

Associative Professor

Department of ECE
BMS College of Engineering

Head of Department

(Dr. K. P. Lakshmi)

Professor and Head
Department of ECE
BMS College of Engineering

(Dr. Bheemsha Arya)

Principal

BMS College of Engineering

External Viva

Name of the Examiners

Signature with Date

- 1.
- 2.

DECLARATION

We, **Rithika P M (1BM22EC198), Aisiri G Aithal(1BM23EC400), Dhananjay L (1BM23EC404), M Adarsha (1BM23EC410)**, hereby declare that the Project Work-2 entitled **A Wearable EMG-Morse Code Silent Communication with AES Security** is a bonafide work and has been carried out by us under the guidance of **Dr. Lalitha S** Associate Professor Department of Electronics and Communication Engineering, BMS College of Engineering, Bengaluru submitted in partial fulfilment of the requirement for completion of PROJECT WORK-2 [23EC7PWPJ2] of Bachelor of Engineering in Electronics and Communication during the academic year 2025-26. The project report has been approved as it satisfies the academic requirements in Electronics and Communication Engineering, Visvesvaraya Technological University, Belagavi, during the academic year 2025-26. We further declare that, to the best of our knowledge and belief, this Project Work-2 has not been submitted either in part or in full to any other university.

Place: Bengaluru

Date:

RITHIKA P M	1BM22EC198
 AISIRI G AITHAL	1BM23EC400
 DHANANJAY L	1BM23EC404
 M ADARSHA	1BM23EC410

ACKNOWLEDGEMENTS

We take this opportunity to express our profound gratitude to the respected principal Dr. Bheemsha Arya, BMS College of Engineering for providing a congenial environment to work in. Our sincere gratitude to Dr. K. P. Lakshmi, Head of the Department, Electronics and Communication Engineering for encouraging and providing opportunity to carry project work 2 in the department.

We heartfully thank our guide Dr. Lalitha S for the guidance and constant encouragement throughout the course of this project work 2 without which this project would not be successful.

A number of personalities, in their own capacities have helped us in carrying out this project work 2. We would like to take this opportunity to thank them all.

Last but not the least, we thank our friends and family for their encouragement and help in accomplishing the objective of the project work 2.

- 1) Rithika P M: 1BM22EC198
- 2) Aisiri G Aithal: 1BM23EC400
- 3) Dhananjay L: 1BM23EC404
- 4) M Adarsha: 1BM23EC410

ABSTRACT

This project presents a novel human-machine communication system that converts muscle contractions into secure, transmittable text. The system uses an EMG sensor to detect voluntary muscle contractions and classifies them as either short or long gestures, which are mapped to Morse-style dot and dash symbols. These symbols accumulate to form individual letters, while a 300ms silence interval is used to identify the end of each letter. A capacitive touch sensor acts as a mode selector; only when the touch sensor remains pressed does the system record and assemble letters. Once the touch is released, the collected letters are mapped into meaningful words.

To ensure secure wireless transmission, the generated word is encrypted using AES and sent via Bluetooth to a Flutter-based mobile application. The mobile app decrypts the message and displays the final word to the user. This workflow enables hands-free, button-free communication for users with motor impairments or limited mobility, while ensuring privacy and data security through cryptographic protection. The system combines physiological sensing, gesture-based encoding, real-time processing, and encrypted wireless communication to provide an efficient, accurate, and secure text-generation interface.

TABLE of CONTENTS

Abstract
List of figures
List of tables
List of abbreviations

<u>TOPIC</u>	<u>Page No</u>
Chapter 1: Introduction	1
Chapter 2: Literature survey	4 13
Chapter 3: Problem Analysis & Solution	
3.1 Problem Definition	
3.2 Proposed Solution	
	17
Chapter 4: Methodology & Implementation	
4.1 Block Diagram	
4.1.1 Pictorial Representation	
4.1.2 Structure of the work	
4.1.3 Flow Chart	
Chapter 5: Results & Discussion	36
Chapter 6: Future Trends	45
6.1 Conclusion	
6.2 Future Trends	
References	49
Plagiarism Report	53

List of figures

Fig. No.	Name	Page. No
4.1	Block diagram	17
4.1.1.1	Pictorial Representation of the model	22
4.1.1.2	No contraction to display space	22
4.1.1.3	Full contraction for morse code	22
4.1.2	Structure of hardware flow	24
4.1.3.1	Software Work Flow	27
4.1.3.2	System flow	31
5.1	Morse code mapping using muscle contraction	37
5.2	Morse code mapped to letter 'A'	38
5.3	Morse code mapped to letter 'D'	38
5.4	Morse code mapped to letter 'R'	39
5.5	Word Detection and Encrypting Output	40
5.6	Word Detection and Encryption for 'SOS'	40
5.7	Word Detection and Encryption for 'HELP'	40
5.8	Decryption in Flutter Application and Display	41

List of tables

Table.No	Description of Table	Page. No
5.1	Burst Classification and Morse Rules	36
5.2	Morse Mapping (Selected Alphabet)	38
5.3	Letter Recognition Statistics (200 Attempts)	39
5.4	AES Encryption Example for the Word	41

List of abbreviations

Abbreviations	Full Form
IEEE	Institute of Electrical and Electronics Engineers
ADC	Analog-to-Digital Converter
AES	Advanced Encryption Standard
BT	Bluetooth
EMG	Electromyography
HC-05	Bluetooth Module used for Serial Communication
OLED	Organic Light Emitting Diode
Li-Po	Lithium-Polymer
STM32	STMicroelectronics 32-bit Microcontroller

Chapter 1:

Introduction

In the modern era, communication stands as one of the most fundamental pillars supporting both everyday civilian life and complex defence operations. Whether it involves simple personal interactions or advanced strategic coordination between military units, the uninterrupted exchange of information determines the efficiency and safety of the people involved. The shift toward digital and wireless communication systems has undoubtedly improved global connectivity, but it has also introduced several vulnerabilities and limitations that cannot be ignored. Systems that rely heavily on network availability, satellite coverage, or radio signals often struggle in environments where connectivity is unstable, restricted, or intentionally disrupted. Furthermore, many conventional communication technologies are prone to interception, making them unsafe for situations that demand confidentiality and resilience against security breaches.

These challenges demonstrate a clear need for alternative communication mechanisms that can operate reliably even when mainstream systems fail. The increasing emphasis on safety, secrecy, mission-critical communication, and user accessibility has motivated researchers to explore unconventional yet effective solutions. One of the most promising directions involves integrating human-machine interaction with bio-signal processing. By focusing on natural signals generated by the human body signals that are always present, require no external network, and can be captured using compact sensors it becomes possible to design communication systems that are portable, secure, and highly adaptable.

Among the various biological signals available, muscle activity stands out as a particularly robust option. The electrical impulses produced during muscle contraction, known as electromyographic (EMG) signals, provide a dependable medium through which human intention can be translated into digital form. These signals can be captured easily using non-invasive sensors and processed in real time to represent meaningful patterns. When mapped to predefined communication codes, muscle signals offer a unique method of transmitting information without relying on speech, visible gestures, or conventional electronic communication channels. This capability opens the door to innovative applications,

especially in situations where communication must be discreet, silent, or independent of external infrastructure.

Morse code, despite being one of the oldest communication systems in existence, remains remarkably relevant even today. Its resilience lies in its simplicity messages are formed using only two basic components: short signals (dots) and long signals (dashes). Because Morse code requires minimal bandwidth and can be generated using sound, light, touch, or electrical impulses, it continues to serve as a reliable method in low-signal or emergency scenarios. When combined with modern signal-processing techniques and secure data-handling protocols, Morse code transforms from an old communication method into a powerful, lightweight, and highly dependable tool. It becomes especially advantageous in environments where digital networks are unavailable, compromised, or too risky to use.

In the system proposed in this work, the key idea is to merge the long-standing reliability of Morse code with the capabilities of modern EMG-based sensing and secure processing methods. The approach focuses on capturing subtle muscle actions particularly voluntary contractions through sensors placed on the user's arm. These signals are then interpreted and converted into Morse-coded messages. This methodology not only minimizes the physical effort required from the user but also ensures that communication can occur silently and discreetly. Such characteristics are vital in defence contexts where stealth, secrecy, and quick communication can be crucial for mission success.

Furthermore, security plays a central role in the system's design. Since any information transmitted in defence or sensitive civilian scenarios must be protected from unauthorized access, the system incorporates secure encoding techniques to ensure confidentiality. Even if a third party attempts to intercept the communication, the encoded data would be incomprehensible without the correct decoding mechanism. This layered approach biological input, code conversion, and secure encryption creates a highly robust communication pipeline suited for diverse applications.

The scalability of the system adds another dimension to its usefulness. Because the basic structure relies on human muscle activity and simple coding schemes, it can be adapted for multiple use cases without significant alteration. For individuals with speech impairments or motor disabilities, this technology offers an accessible way to communicate through

minor voluntary movements. For field personnel working in remote or hazardous environments, it ensures that essential information can be transmitted even when conventional devices fail. In defence operations, it provides a silent and secure channel for transmitting commands or alerts without revealing the user's position.

Chapter 2:

Literature survey

1. Morse Code Representation using EMG Signals – JeHun Gu, Mansung Park, Kimoon Kang, and Hyun-Chool Shin (2019)

The work by JeHun Gu and colleagues (2019) presents an innovative approach to generating Morse code using electromyographic (EMG) signals derived from voluntary muscle activity. In their system, simple physical actions such as fist folding and stretching are captured through EMG sensors. These signals are then processed and categorized based on their duration and intensity. Short muscle contractions are interpreted as Morse “dots,” while longer contractions are mapped as “dashes.” Through this mapping, users can produce complete Morse sequences, which are subsequently translated into meaningful text messages.

A central contribution of this research lies in demonstrating that EMG signals can serve as a dependable medium for communication, especially for individuals unable to speak or perform traditional gestures. Because the system relies on minimal physical movement, it becomes highly suitable for users with limited mobility due to neurological or muscular disorders. Even individuals with severe speech impairments can communicate efficiently by generating controlled muscle contractions, bypassing the need for speech, handwriting, or complex gesture-based systems.

The authors place strong emphasis on signal accuracy and reliability. Their study reports high recognition rates, indicating that the classification algorithm can precisely distinguish between short and long contractions despite inherent noise in biological signals. This reliability makes the system practical for real-time communication, where misinterpretation of signals could disrupt message clarity.

Another significant aspect of the study is its focus on user comfort and accessibility. EMG sensors can be worn comfortably on the forearm and require no invasive setup, making the system both user-friendly and non-intrusive. This design advantage enhances long-term usability in assistive technology scenarios, where continuous or repeated use is expected.

The research also highlights the broader implications of EMG-based Morse communication.

Since Morse code is universally recognized and requires minimal computational resources, the authors' method can be applied not only in assistive medical technologies but also in low-bandwidth communication systems, emergency communication, and environments where silent or discreet messaging is necessary.

2. Human–Robot Interaction: A Myo Armband Using EMG and IMU Signals – Uzair Muhammad et al. (2020)

The study by Uzair Muhammad and colleagues (2020) explores the Myo armband as an affordable and intuitive interface for human–robot interaction. The Myo armband is a wearable device equipped with both electromyography (EMG) sensors and an inertial measurement unit (IMU). This combination allows the system to capture muscle activity along with motion-related data such as acceleration, orientation, and angular velocity. By merging these two types of signals, the researchers were able to create a gesture-recognition system capable of controlling robotic platforms with high precision and responsiveness.

One of the key motivations behind this work is the limitation of traditional robotic control interfaces. Conventional methods often rely on bulky hardware, complex controllers, or high-cost devices that are not user-friendly for non-experts. In contrast, the Myo armband offers a natural, low-cost alternative by enabling users to interact with robots through simple hand and arm movements. This makes the system highly accessible for everyday users and reduces the learning curve typically associated with robot operation.

The study demonstrates how specific gestures detected from EMG patterns such as muscle contractions, hand posture changes, and wrist movements can be mapped to robot commands. The IMU data further enhances accuracy by providing contextual information on limb orientation, ensuring the system can differentiate between similar muscle patterns occurring in different physical positions. This multimodal sensing approach significantly improves gesture recognition reliability.

Beyond basic robot control, the research highlights several meaningful applications. In rehabilitation environments, for example, patients recovering from motor impairments can use the Myo armband to interact with training robots, enabling personalized and interactive therapy sessions. Similarly, in human–computer interaction, the system allows users to control digital interfaces without needing keyboards, mice, or touchscreens, benefiting

individuals with mobility challenges.

The study concludes that the Myo armband offers a promising pathway toward intuitive and natural human–robot interfaces. Its affordability, portability, and ease of integration make it suitable for a wide range of real-world applications. By leveraging wearable sensor technology, the researchers demonstrate that robots can be controlled in a more human-centered and accessible manner, paving the way for smarter and more inclusive interaction systems.

3. A Wearable Device Based on IMU and EMG Sensors for Remote Monitoring of Elbow Rehabilitation – Diego M. Lavado and Emir A. Vela (2022)

The study by Diego M. Lavado and Emir A. Vela (2022) presents a wearable system that combines inertial measurement unit (IMU) sensors with electromyography (EMG) sensors to monitor elbow rehabilitation remotely. Their work addresses a major challenge in physical therapy: patients often require continuous supervision and feedback to ensure that rehabilitation exercises are performed correctly. However, frequent visits to clinics are not always possible due to factors such as travel limitations, healthcare costs, and patient availability. The wearable device proposed in this study offers an effective solution by enabling real-time monitoring outside clinical environments.

The system integrates IMU sensors to capture the movement of the elbow joint, providing detailed information on arm orientation, angle changes, and motion dynamics. At the same time, EMG sensors record muscle activity, offering insights into the level of muscular engagement during exercises. Together, these sensors create a comprehensive picture of the patient's rehabilitation performance. By analyzing both motion and muscle activity, healthcare professionals can assess not only whether the movement is correct but also whether the appropriate muscles are being activated.

A significant contribution of this research is the wireless data transmission capability. The device sends sensor data to a remote monitoring platform, allowing physiotherapists to observe patient progress in real time. This eliminates the need for constant in-person supervision and enables doctors to intervene early if improper technique or abnormal recovery patterns are detected. Such timely guidance can greatly improve rehabilitation outcomes and reduce the risk of repeated injury or delayed healing.

Another notable aspect of the study is its focus on user comfort and practicality. The wearable device is lightweight and designed for easy attachment to the arm, ensuring that patients can use it regularly without discomfort. Its portability allows patients to perform rehabilitation exercises from home while still receiving professional oversight. This flexibility is especially beneficial for individuals recovering from surgeries, sports injuries, or neurological conditions affecting arm mobility.

The authors also highlight the potential of their system to support personalized rehabilitation plans. By analyzing long-term data collected from the sensors, healthcare providers can track improvements, adjust exercise difficulty, and tailor therapy according to each patient's progress. This personalized approach enhances the effectiveness of treatment and promotes better engagement from the patient.

4. Hand Movement Detection Device Using IMU and sEMG – Niti Petranon and Sumet Umchid (2023)

The work of Niti Petranon and Sumet Umchid (2023) presents a prototype device designed to accurately detect and interpret hand movements using a combination of inertial measurement unit (IMU) sensors and surface electromyography (sEMG) sensors. Their research addresses a significant communication challenge faced by hearing-impaired individuals: the difficulty of interacting with people who do not understand sign language. By developing a system capable of converting sign language gestures into readable text, the authors aim to create a more inclusive communication environment.

The device operates by capturing two distinct types of data. IMU sensors track hand orientation, acceleration, and spatial movement patterns, while sEMG sensors measure the electrical activity produced by the muscles of the hand and forearm during gesture formation. This multimodal sensing strategy is a key strength of the system, as combining motion data with muscle activity allows for more reliable gesture classification. Many hand gestures involve subtle finger movements or involve similar hand shapes but different motion trajectories, and using both sensor types helps the system distinguish these differences with higher accuracy.

The authors highlight the importance of real-time recognition in practical communication scenarios. Their prototype processes sensor data on the fly, enabling immediate translation

of gestures into text. This capability makes the system suitable for dynamic conversations, where delays in recognition could disrupt communication flow. The output is displayed in text format, allowing users unfamiliar with sign language to understand messages without any prior training.

Another significant aspect of this research is its potential impact on assistive technologies. Traditional sign language recognition systems often rely on cameras, which can be sensitive to lighting conditions, background clutter, and occlusions. In contrast, wearable sensor-based systems like the one proposed in this study offer greater robustness and portability. They do not depend on environmental conditions and allow users to move freely without needing a fixed camera setup. This makes the device highly practical for daily use.

The study also emphasizes that combining IMU and sEMG sensors greatly enhances system accuracy compared to using either sensor alone. Multimodal data fusion reduces misclassification errors and helps the system handle variations in gesture speed, hand position, and individual muscle characteristics. These improvements result in a more reliable communication aid that can perform consistently in real-world conditions.

5. EMG Dataset for Arabic Sign Language – Amina B. H. Amor, O. El Ghouli, and M. Jemni (2023)

The study by Amina B. H. Amor, O. El Ghouli, and M. Jemni (2023) presents a valuable contribution to the field of gesture recognition by introducing a dedicated surface EMG dataset for Arabic Sign Language. The dataset was collected using the Myo Armband, a wearable device equipped with multiple EMG sensors that capture muscle activity from the forearm. This work addresses the critical need for region-specific datasets to support machine learning research, especially in communities where language and gesture structures differ from widely studied sign languages like American Sign Language (ASL).

The authors highlight that sign language interpretation systems heavily depend on high-quality datasets to train accurate and reliable models. However, Arabic Sign Language has been underrepresented in technological research due to the lack of publicly available datasets. By creating and publishing this EMG-based dataset, the researchers fill an important gap, enabling developers and scientists to build recognition systems tailored specifically for Arabic-speaking users.

The dataset captures muscle activity patterns corresponding to various gestures and hand movements used in Arabic Sign Language. Because EMG signals reflect internal muscular effort rather than external appearance, they offer advantages over camera-based datasets. EMG-based datasets eliminate issues such as variations in lighting, background distractions, or differences in hand appearance. This makes the dataset particularly useful for creating highly robust gesture-recognition models that perform consistently across different environments and users.

Another key strength of this dataset lies in its suitability for mobile and embedded applications. Since the Myo Armband is compact and wireless, the recorded signals closely represent real-world usage scenarios where users rely on portable devices. This makes the dataset ideal for developing lightweight models that can run efficiently on smartphones, microcontrollers, or wearable platforms. As the demand for on-device processing increases, such datasets become essential for advancing assistive technologies.

The authors also emphasize the broader impact of their contribution. By enabling machine learning models to recognize Arabic Sign Language through EMG signals, the dataset supports the development of communication tools for individuals with hearing or speech impairments. These tools can help bridge communication gaps in educational settings, workplaces, public services, and daily interactions. In addition, researchers can use the dataset to study muscle activity patterns associated with different gestures, contributing to deeper understanding of biomechanical and linguistic aspects of sign languages.

6. High Speed and Accuracy of Encryption–Decryption Techniques Using Morse Code – Pulkit Singh et al. (2024)

The study by Pulkit Singh and colleagues (2024) presents an innovative approach to secure communication by integrating Morse code with modern microcontroller technology. Their work focuses on designing an Arduino-based system capable of encrypting text into Morse code and subsequently decrypting Morse sequences back into readable text. The central objective of the research is to achieve a communication method that is not only secure but also reliable, fast, and computationally lightweight.

The authors revisit Morse code—one of the earliest forms of digital communication—and demonstrate how it can still be highly effective in contemporary systems when paired with

microcontrollers. Morse code's simplicity, consisting only of dots and dashes, enables fast processing and minimal resource usage. This makes it an attractive alternative for low-power embedded devices that require secure information exchange without the overhead of complex cryptographic algorithms.

In their system, input text is first converted into Morse code using a predefined mapping of characters. The Arduino then transmits this encoded message, which can be sent via LED blinking, sound pulses, or digital signals depending on the application. For decryption, the system detects the incoming Morse signals, interprets the timing of short and long pulses, and reconstructs the original text. The researchers highlight that achieving precise timing detection is crucial, as it ensures high accuracy during decoding.

One of the major contributions of this work is its emphasis on speed and reliability. The system is optimized to process Morse encoding and decoding rapidly, enabling near real-time communication. The authors also report high accuracy rates, illustrating that even simple embedded platforms like Arduino can handle structured encoding schemes efficiently. This makes the technique suitable for scenarios where quick and dependable communication is essential.

The study further explores security implications. While Morse code alone does not provide strong cryptographic protection, embedding it within controlled communication channels or layering it with additional encryption can significantly enhance security. The structured nature of Morse code also makes it easier to apply further transformations such as obfuscation or key-based pattern variation, enabling lightweight cryptography for resource-constrained environments.

Pulkit Singh and colleagues also discuss practical applications of their system. It can be used in military communication for sending short, encrypted messages silently through light signals, in disaster recovery operations where conventional communication networks fail, or in IoT systems requiring low-complexity secure messaging. The system's portability and low cost further support its usability across diverse fields.

7. Application of AES Algorithm in Intelligent Wireless Network Information Security Protection System – Qiyu Chen (2024)

The study by Qiyu Chen (2024) focuses on strengthening information security in intelligent

wireless networks, which face growing risks due to rapid expansion of IoT devices and next-generation communication technologies. As wireless systems become more interconnected and data-intensive, protecting the confidentiality, integrity, and authenticity of transmitted information becomes increasingly challenging. Chen's work proposes a comprehensive security framework built around the Advanced Encryption Standard (AES) algorithm, complemented by authentication and intrusion detection mechanisms.

AES, a widely adopted symmetric encryption algorithm, is recognized for its strong resistance to cryptographic attacks, high processing efficiency, and suitability for both hardware and software implementations. In this research, the author leverages AES to secure data during wireless transmission, ensuring that unauthorized users cannot intercept or manipulate sensitive information. The study emphasizes that AES's balance of speed and security makes it ideal for intelligent wireless environments, where real-time communication and resource constraints must be considered.

In addition to encryption, Chen integrates authentication techniques into the framework to verify the identity of communicating devices. This step is vital in preventing unauthorized access, especially in IoT networks where large numbers of heterogeneous devices communicate autonomously. By ensuring that only legitimate nodes can participate in the network, the system reduces vulnerabilities related to spoofing and impersonation attacks.

A further enhancement to the framework is the inclusion of intrusion detection mechanisms. These mechanisms monitor network behavior, analyze traffic patterns, and identify anomalies or suspicious activities that may indicate ongoing attacks. This layered security approach — encryption for confidentiality, authentication for access control, and intrusion detection for threat monitoring — provides comprehensive protection suitable for complex wireless architectures.

The author also highlights practical considerations for next-generation networks such as 5G and intelligent IoT ecosystems. These networks demand high-speed data exchange, low latency, and the ability to process large volumes of information simultaneously. Chen shows that AES-based security can be optimized for these demands through lightweight implementations, parallel processing, and hardware acceleration techniques. As a result, the framework can maintain strong security without compromising network performance.

Applications of the proposed system extend across various sectors, including smart homes, industrial IoT systems, healthcare monitoring, autonomous vehicles, and military communication networks. In each of these environments, secure wireless communication is essential to ensure safety, reliability, and privacy. Chen's study demonstrates that embedding AES into wireless network architectures provides a scalable and future-ready solution to evolving cybersecurity threats.

Chapter 3:

Problem Analysis & Solution

3.1 Problem Definition

In modern communication environments—ranging from everyday personal interactions to highly sensitive military and industrial operations—the need for secure, reliable, and discreet information exchange is becoming increasingly critical. Although numerous communication technologies exist today, many of them fall short when applied to high-risk or constrained environments. Conventional systems depend heavily on stable external infrastructure such as camera setups, network connectivity, or complex gesture-recognition devices. These systems often prove to be impractical in dynamic, unpredictable, or hostile conditions where reliability and security cannot be compromised.

One of the major limitations is the reliance on camera-based or vision-based gesture recognition systems. While these systems are widely used for human-machine interaction, they suffer from several inherent drawbacks. Their accuracy is significantly affected by lighting conditions, camera placement, background clutter, and occlusion of hand movements. Even in controlled environments, gestures that look visually similar can be mistakenly classified, causing communication delays, errors, and user frustration. These limitations make vision-based systems unsuitable for mission-critical applications where even a small misinterpretation can lead to serious consequences.

Beyond accuracy issues, many existing communication methods fail to address another major threat: data security. As wireless communication grows, so does the risk of interception, eavesdropping, and data manipulation. Systems that do not incorporate strong encryption or secure transmission protocols leave sensitive information vulnerable. In military operations, this could compromise mission plans or put personnel at risk. In industrial communication, it could result in unauthorized access to confidential data. Even for personal use, unsecured communication exposes users to privacy breaches and identity misuse. The absence of built-in security mechanisms in many assistive or gesture-based communication systems represents a significant gap that must be addressed.

Another major challenge arises in the context of accessibility for individuals with physical or speech impairments. Many of the current communication tools require large motions,

precise gestures, or visual interpretation, which are not feasible for people with limited mobility. Speech-based systems are also impractical for individuals with speech disabilities. While some sensor-based solutions exist, they are often bulky, expensive, or difficult to operate independently. As a result, individuals who rely on assistive communication devices frequently face barriers in conveying messages quickly, conveniently, and confidentially.

Furthermore, many existing systems lack portability and ease of deployment. Devices that require cameras, fixed installations, or complex hardware setups cannot be used seamlessly in outdoor environments, remote locations, or emergency situations. In contrast, a wearable, body-signal-based communication system would provide a practical alternative by allowing users to communicate silently and securely regardless of their surroundings.

In high-stakes environments such as defence, the importance of silent communication cannot be overstated. Traditional communication—whether spoken, typed, or gesture-based—may give away the user’s position or may not be feasible in covert operations. Soldiers or responders may need to transmit critical messages without exposing themselves visually or audibly. Existing tools do not fully support this requirement, leaving a gap for technologies that enable silent, muscle-based, encrypted communication.

All these challenges point to a clear need for an alternative communication system that is accurate, compact, secure, and independent of visual cues or environmental factors. Such a system must:

- Reliably interpret human intent using signals that are always available (e.g., muscle activity)
- Operate without reliance on cameras, speech, or large gestures
- Ensure confidentiality through strong encryption
- Function effectively even in low-light, noisy, or network-limited environments
- Be affordable, wearable, and accessible to individuals with disabilities
- Support real-time communication without delays or misclassification

The lack of a communication tool that satisfies all these criteria creates a significant technological gap. Individuals in defence roles, industrial settings, medical contexts, and

assistive communication scenarios would benefit greatly from a system that enables silent, secure, and highly accurate message transmission using natural human signals.

3.2 Proposed Solution

To address the limitations of current communication systems—particularly their dependence on cameras, speech, or large gestures—this project proposes a compact, wearable, and secure communication method based on Electromyography (EMG) signals. The system focuses on capturing subtle muscle activity from the forearm, translating these signals into Morse code, and transmitting them securely using modern encryption and wireless technology. This approach ensures silent, reliable, and confidential communication, suitable for assistive applications as well as high-security environments.

The process begins with an EMG sensor placed on the user's forearm. Whenever the user performs a voluntary muscle contraction, the sensor detects the electrical activity generated by the muscle fibres. These signals are processed to differentiate between short and long contractions. The system then maps these contractions directly to Morse code, where short signals represent dots and longer signals represent dashes. This method allows users to form words and messages using simple muscle actions, eliminating the need for visual interpretation or audio input.

Once the Morse code has been generated, the system converts the sequence into alphanumeric text and then into binary format to prepare it for digital encryption. To ensure privacy and prevent unauthorized access, the binary message is encrypted using the Advanced Encryption Standard (AES). AES provides strong protection against data interception and is widely used in secure communication systems due to its speed, robustness, and resistance to modern cryptographic attacks. By integrating AES into the communication pipeline, the system guarantees that even if the signal is intercepted, the message remains unreadable without the correct decryption key.

After encryption, the secure data is transmitted wirelessly to a mobile device using Bluetooth communication. Bluetooth offers low power consumption, portability, and universal compatibility, making it ideal for wearable applications. It enables communication even in environments without network connectivity, increasing the practicality of the system for

field use, emergency situations, or remote locations.

On the receiving end, a dedicated mobile application receives the encrypted data through Bluetooth. The app performs AES decryption using the shared key, converts the binary back into readable text, and displays the final message to the user. This provides a seamless and user-friendly interface that requires minimal technical knowledge and ensures quick message interpretation.

The proposed solution provides several benefits. It allows silent and discreet communication, which is essential in military operations, rescue missions, or situations where noise must be minimized. It also offers an accessible communication method for individuals with speech or motor impairments, enabling them to express themselves using only small muscle movements. Additionally, the integration of AES encryption ensures strong message confidentiality, making the system suitable for secure personal use or sensitive industrial environments.

Overall, this solution brings together EMG-based input, Morse coding, cryptographic security, and wireless communication into a unified, practical system. By leveraging natural muscle signals, it bypasses the limitations of vision-based or speech-based systems and provides a modern, reliable method for real-time communication in diverse applications.

Chapter 4:

Methodology & Implementation

4.1 Block Diagram

The proposed system is designed as a compact, portable, and wearable communication device capable of capturing human muscle activity and motion signals, processing them intelligently, and transmitting the interpreted information wirelessly to a mobile phone. The complete architecture is divided into two major sections:

1. **Transmitter Part (Wearable Unit)**
2. **Receiver Part (Mobile Phone Application)**

Together, these units form an efficient, real-time communication system that can be used for secure messaging, assistive communication, gesture recognition, and silent command transmission. The block diagram illustrates how each module interacts within the system to achieve continuous signal acquisition, processing, encoding, encryption, and wireless data transfer.

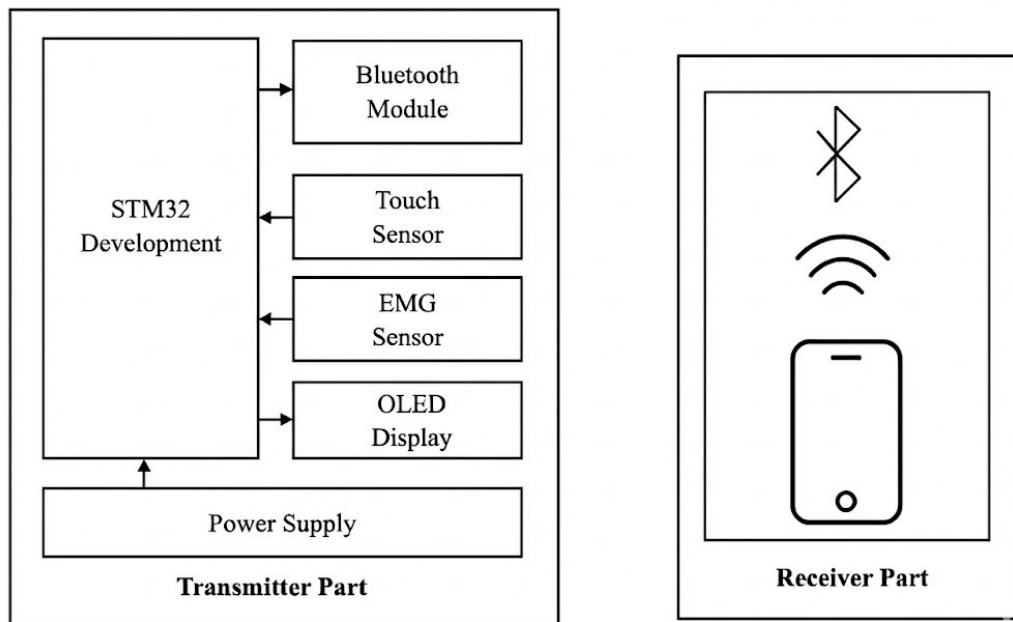


Fig. 4.1: Block Diagram

1. Transmitter Part (Wearable Unit)

a. Power Supply

The power supply is responsible for delivering stable and continuous electrical energy to all components in the wearable transmitter unit. A rechargeable Li-ion or Li-Po battery is typically used because it offers high energy density and keeps the device lightweight and portable, which is essential for wearable applications. The battery output is regulated through a voltage control circuit to ensure that the STM32 microcontroller, EMG sensor, IMU sensor, and Bluetooth module receive a consistent and safe operating voltage. Without this regulation, sudden fluctuations could disrupt sensor readings or processing accuracy. By providing clean and reliable power, the power supply enables uninterrupted operation of the system and ensures long-term usability during daily wear.

b. EMG Sensor (Electromyography Sensor)

The EMG sensor serves as the primary input interface for detecting the user's muscle activity. Positioned on the forearm or another suitable muscle group, it captures the tiny electrical signals produced during muscle contractions. Since these signals are very small and often contain noise, the sensor includes basic amplification and filtering before sending the signal to the STM32 microcontroller for digital processing. The EMG data reflects the user's intentional muscle movements, which allows the system to interpret short and long contractions for Morse code generation or command control. Thus, the EMG sensor plays a crucial role in converting biological muscle patterns into meaningful digital information.

c. Touch Sensor (Mode Switching and Message Control)

The touch sensor in the wearable unit is not used for gesture recognition but functions as a **control interface** for user interaction. It allows the user to switch between different operating modes and to trigger specific actions during the communication process. The system includes multiple modes such as **Typing Mode**, **Encryption Mode**, and **Sending Mode**, and the touch sensor enables smooth transitions between these states. For example, a single tap may activate typing mode, allowing EMG inputs to be interpreted as Morse code characters. A long press may switch the device into encryption mode, where the typed

message is secured using AES. Another tap could send the encrypted message to the mobile device via Bluetooth.

By assigning simple touch interactions for mode control, the system becomes far more intuitive and user-friendly. The touch sensor also eliminates the need for additional buttons or hardware, reducing the size and complexity of the wearable device. This makes the overall system lightweight, efficient, and easy to operate—especially for users with limited mobility or in situations where quick and silent communication is essential.

d. STM32 Development Board (Microcontroller Unit)

The STM32 microcontroller acts as the central processing node of the transmitter and is responsible for handling all computational tasks. It begins by digitizing the analog EMG and IMU signals through its built-in ADC channels. After digitization, the STM32 applies signal processing techniques such as filtering, threshold detection, and duration measurement to identify meaningful features from the data. The microcontroller then classifies muscle contractions into short or long pulses and maps them into corresponding Morse code symbols. These symbols may be further translated into ASCII characters or words. If secure communication is required, the STM32 encrypts the processed message using AES algorithms before sending it to the Bluetooth module. Through these operations, the STM32 ensures accurate interpretation and efficient transformation of sensor data into usable communication output.

e. Bluetooth Module (Wireless Transmitter)

The Bluetooth module is responsible for wirelessly transmitting the processed data from the STM32 microcontroller to the mobile receiver. It provides a low-power and short-range communication link, making it ideal for wearable applications that require mobility and convenience. The module ensures stable and real-time data transfer, enabling immediate interpretation on the receiving device. Depending on system configuration, the transmitted data may include raw sensor values, Morse-coded information, or encrypted binary packets for secure communication. By removing the need for wired connections, the Bluetooth module enhances comfort and usability while maintaining efficient, continuous communication with the mobile application.

f. OLED Display (Real-Time Feedback Unit)

The OLED display provides real-time visual feedback to the user about the current operation and system status. It shows the selected mode (Typing, Encryption, Sending), displays the Morse characters being generated, and confirms when a message is successfully encrypted or transmitted. This visual interface greatly enhances usability by allowing the user to verify their input before sending. The OLED consumes minimal power, offers high contrast, and remains readable even in low-light environments, making it ideal for wearable devices. By integrating the OLED display, the system becomes more interactive, user-friendly, and reliable for continuous communication tasks.

2. Receiver Part (Mobile Device)

a. Bluetooth Receiver (Smartphone Interface)

The smartphone operates as the central receiving unit for all data transmitted by the wearable device. Using its integrated Bluetooth hardware, the mobile device establishes a secure wireless link with the transmitter. Once paired, the device remains in an active listening mode, continuously receiving data packets in real time. This includes Morse-coded messages, binary information, or encrypted AES packets depending on the system's operational mode. Bluetooth communication is ideal for this application because it provides low-power consumption, stable short-range connectivity, and rapid data exchange without requiring external network infrastructure. This ensures that communication remains uninterrupted even in remote environments, low-signal areas, or indoor conditions. By removing the need for wired connections, the receiver interface enhances mobility, comfort, and practicality for everyday use.

In addition to simply receiving data, the Bluetooth interface also manages connection integrity, error checking, and packet reassembly. If any data packet is lost or corrupted during transmission, the interface can request re-transmission or error correction through built-in Bluetooth protocols. This improves the reliability of the system and ensures that messages remain accurate when displayed to the user. The smartphone thus acts not only as

a communication endpoint but also as a robust gateway for handling secure and reliable data transfer.

b. Flutter-Based Mobile Application (Decoding, Decryption, and Display)

The mobile application, developed using the **Flutter framework**, plays a critical role in processing the received data and converting it into meaningful output. Flutter is chosen because it enables the application to run efficiently on both Android and iOS devices with a single codebase, significantly reducing development effort and ensuring consistent performance across platforms. The app listens for incoming Bluetooth data streams, buffers the packets, and initiates appropriate processing pipelines based on the type of data received.

When the transmitted message is encrypted, the first step performed by the app is **AES decryption**. Using the predefined encryption key shared with the transmitter system, the app decrypts the binary data packet to retrieve the original Morse-coded or text message. AES decryption ensures that only an authorized device with the correct key can interpret the message, protecting the communication from interception or unauthorized access. After decryption, the app decodes Morse sequences back into readable text using an internal lookup table or algorithm.

Flutter's powerful UI capabilities are used to build a clear, intuitive, and responsive interface. The decoded message is displayed in real time, ensuring that the user receives immediate feedback. The application supports smooth transitions, large readable fonts, and customizable themes to make the display accessible for users with different needs. Since Flutter supports reactive programming through widgets, the interface updates instantly whenever new data arrives, enhancing user experience.

Additionally, the application can log past messages for future review, analyze communication patterns, and export logs if required for medical or analytical purposes. Advanced features such as message timestamps, encryption status indicators, connection health monitoring, and error notifications can also be integrated seamlessly due to Flutter's modular architecture.

Beyond basic display functions, the Flutter app acts as the intelligent endpoint of the system—handling secure decryption, decoding sensor-derived data, providing a user-friendly communication interface, and maintaining a stable Bluetooth connection. This makes the receiver unit not just a display device but a powerful processing hub for the entire communication framework.

4.1.1 Pictorial Representation

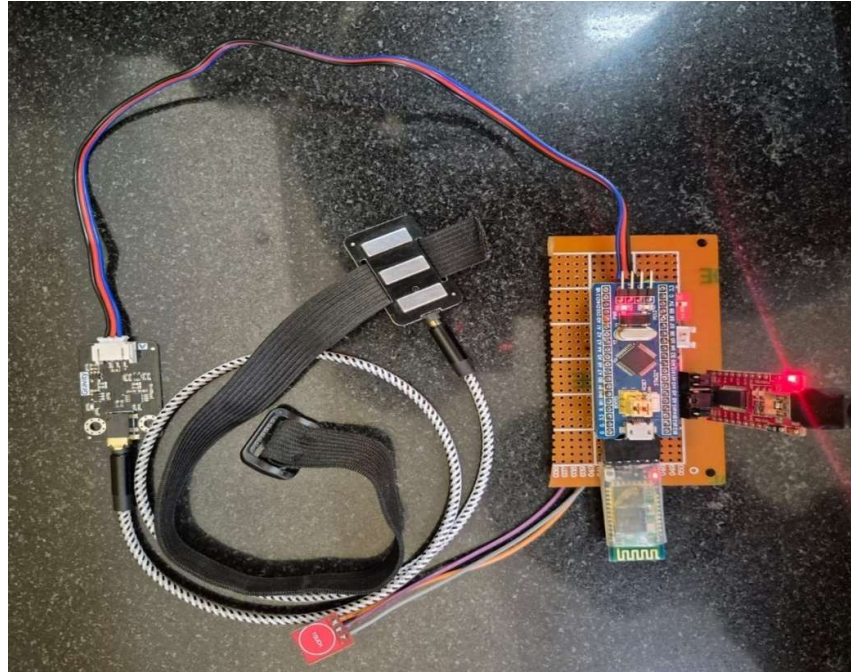


Fig 4.1.1.1 Pictorial Representation of the model



Fig 4.1.1.2 No contraction to display space

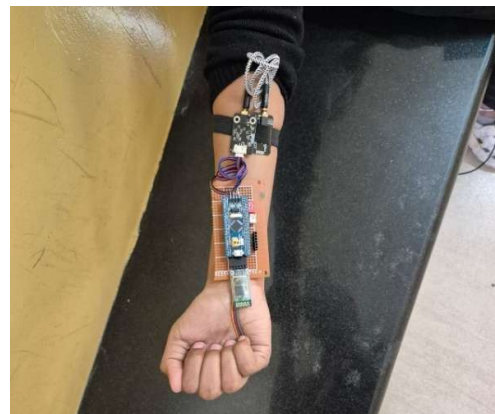


Fig 4.1.1.3 Full contraction for morse code

4.1.2 Structure

The structure of the proposed Secure EMG-Based Morse Communication System is organized into three major layers: Hardware Architecture, Communication Topology, and Software Functional Flow. These layers work together to enable muscle-based input, mode switching via touch, real-time processing, AES-encrypted transmission, and custom mobile-side decoding through a Flutter-developed APK.

1. Hardware Architecture

The hardware layer is composed of two primary units—a wearable transmitter and a mobile receiver—each designed to perform specific roles within the communication system. The wearable transmitter integrates multiple embedded components that collectively handle muscle signal acquisition, mode selection, local processing, visual feedback, encryption, and wireless data transmission. On the other end, the mobile receiver functions as the endpoint for secure message reception and decryption. Together, these subsystems ensure seamless operation by capturing the user's input, processing it into meaningful information, securely transmitting it, and finally presenting the decoded message through an intuitive mobile interface. Their combined functionality establishes a complete, reliable, and secure communication pathway from the user's muscle activity to the final displayed output.

- STM32 Microcontroller (Central Unit)

The STM32 microcontroller acts as the main controller of the wearable device. It collects data from the EMG sensor and touch sensor, performs ADC conversion, interprets Morse code, maps characters, and executes AES encryption. It also updates the OLED display and manages the Bluetooth transmission.

- EMG Sensor (Muscle Activity Input)

The EMG sensor detects the electrical activity generated by the user's muscle contractions. These signals form the primary input for Morse code generation, where short contractions represent dots and long contractions represent dashes.

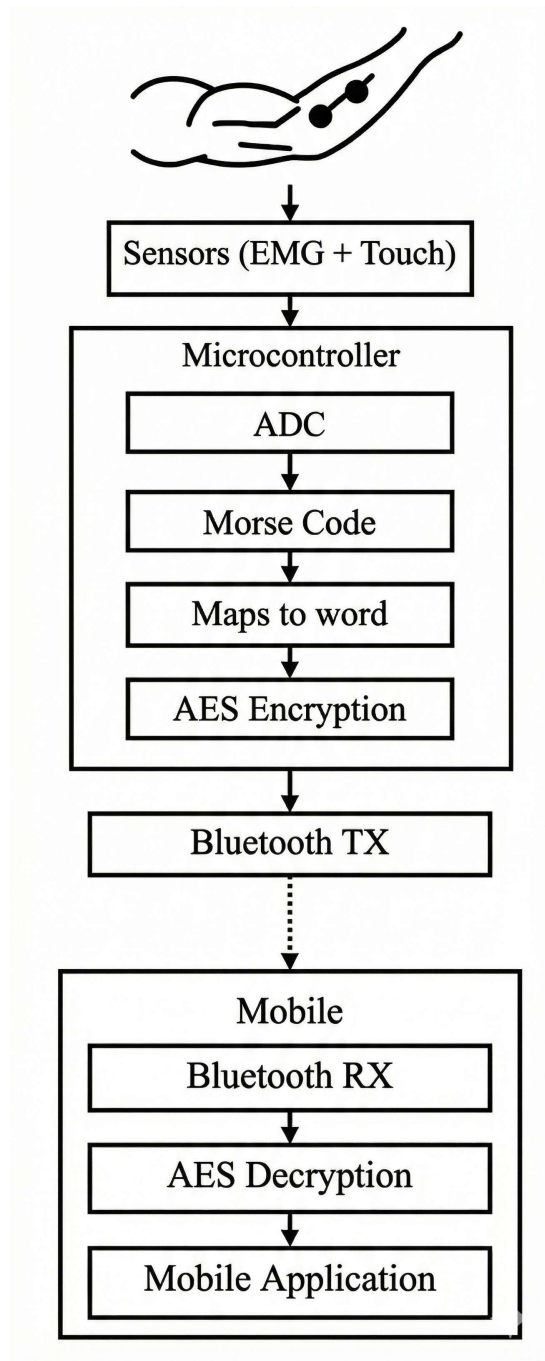


Fig 4.1.2 Structure of hardware flow

- Touch Sensor (Mode Switching and Control)

The touch sensor functions as the system's control interface. It allows the user to switch between different operation modes such as Typing Mode, Encryption Mode, and Sending Mode. It is also used to confirm actions such as sending the typed message. This makes the

system intuitive and eliminates the need for physical buttons.

- OLED Display Module

The OLED display provides real-time visual feedback to the user. It shows the current mode, the detected Morse characters, encryption confirmation, and message-sending status. This improves usability by allowing the user to verify the message before transmission.

- Bluetooth Module (Wireless Transmitter)

The Bluetooth module transmits the encrypted message wirelessly to the mobile application. It supports stable, low-power, short-range communication suitable for wearable systems.

- Power Supply

A rechargeable Li-ion/Li-Po battery powers all components. A voltage regulator ensures stable operation across the STM32, sensors, OLED display, and Bluetooth module. The power unit is optimized for portability and long battery life.

2. Communication Topology

The system uses a point-to-point Bluetooth communication model:

- Wearable Unit : Bluetooth Transmitter

The STM32 sends encrypted data packets through the Bluetooth module.

- Mobile Phone : Bluetooth Receiver

The smartphone receives these packets via Bluetooth using the custom Flutter APK.

- Secure, Low-Latency Data Transfer

The Bluetooth connection ensures:

- Reliable short-range communication
- Low power usage
- Real-time message transfer
- No dependence on internet or external infrastructure

This topology is simple, efficient, and ideal for personal wearable systems.

3. Software Functional Flow

The software layer includes firmware running on the STM32 and the custom Android APK developed by our team using Flutter. The complete flow is as follows:

1. Initialization Phase

- STM32 initializes ADC, GPIOs, OLED, Bluetooth module, and EMG/touch interfaces.
- Default mode is set to Typing Mode.
- The Flutter APK initializes its Bluetooth connection and waits for incoming data.

2. Input Acquisition and Processing Phase

- The EMG sensor captures muscle activity and sends it to the STM32 ADC.
- The microcontroller identifies dots and dashes based on signal duration.
- Touch sensor presses switch between Typing, Encryption, and Sending modes.
- The OLED displays typed characters and mode changes in real time.

3. Mapping and Encryption Phase

- Morse characters are mapped to words or ASCII letters.
- When Encryption Mode is selected via the touch sensor, STM32 performs AES encryption on the message.
- OLED displays confirmation of successful encryption.

4. Transmission Phase

- The encrypted data is sent via Bluetooth to the smartphone.
- OLED shows "Sending..." and then confirms successful transmission.

5. Reception and Decryption Phase (Flutter APK)

- The Flutter-developed APK receives encrypted packets through Bluetooth.
- AES decryption is performed within the app using the shared key.

- The app displays the final message on-screen in a clean, user-friendly interface.
- Users can view message history or export data if needed.

4.1.3 Flow Chart

A. Software Working Flow:

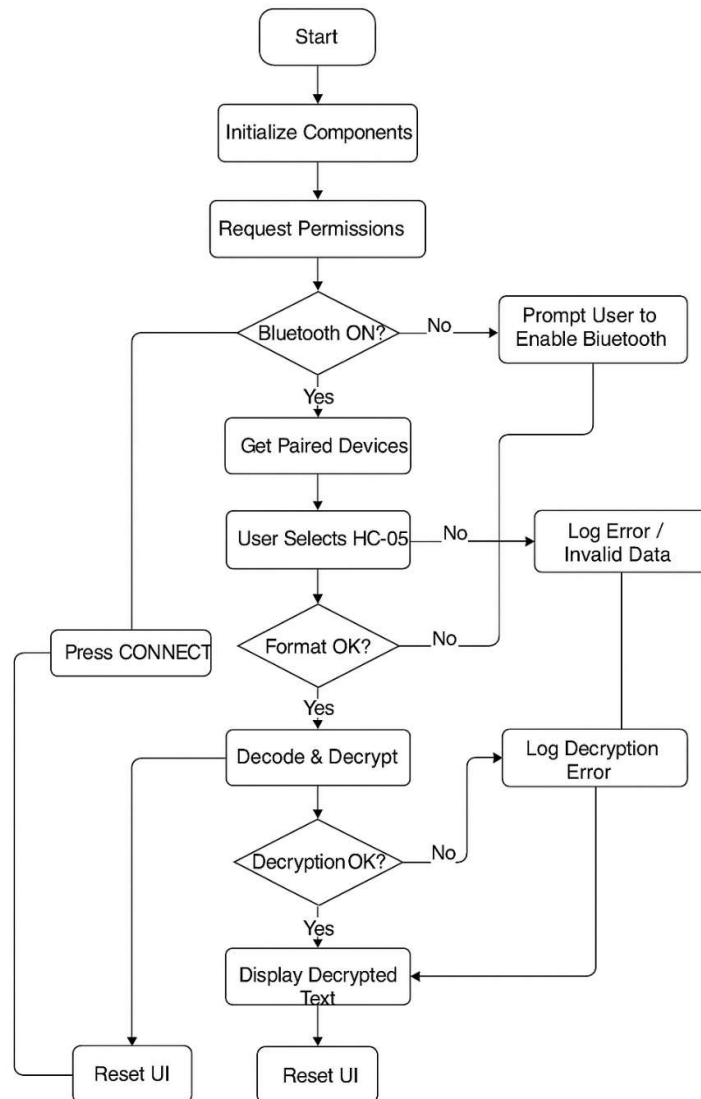


Fig 4.1.3.1 Software Work Flow

1. Start Block

The system begins at the Start point. This block simply marks the initiation of the application workflow. No processing occurs here, but it leads directly to component initialization.

2. Initialize Components

In this block, the application prepares all required internal modules. This includes:

- Initializing Bluetooth adapters
- Initializing the decryption engine (AES)
- Setting up the user interface
- Preparing streams/listeners for incoming data

This step ensures the environment is ready before the system moves to permissions or communication tasks.

3. Request Permissions

Mobile platforms require runtime permissions for Bluetooth operations. During this block, the system requests:

- Bluetooth scan permissions
- Bluetooth connect permissions
- Location permissions (needed for Bluetooth LE scanning)

The application cannot proceed without these permissions, making this a mandatory phase.

4. Bluetooth ON? (Decision Block)

Here, the system checks whether Bluetooth is currently enabled on the device.

If Bluetooth is OFF (No):

The flow moves to the Prompt User to Enable Bluetooth block. After the user enables Bluetooth, the flow returns to this decision point again.

If Bluetooth is ON (Yes):

The flow continues forward to retrieve paired devices.

5. Prompt User to Enable Bluetooth (Action Block)

If Bluetooth is disabled, the system displays a prompt asking the user to turn it on. Once enabled, the system loops back to the Bluetooth ON? decision block.

6. Get Paired Devices

The application retrieves all Bluetooth devices that the phone has paired with earlier. This is necessary because users cannot connect to unpaired devices without pairing first.

7. User Selects HC-05

From the list of paired devices, the user selects the HC-05 Bluetooth module. This module is responsible for sending encrypted messages to the application.

If the user selects incorrectly (No):

The flow moves to Log Error / Invalid Data, then loops back.

If the user selects correctly (Yes):

The process continues to the connection stage.

8. Log Error / Invalid Data (Error Block)

This block records errors when:

- The user selects the wrong device
- The system receives invalid data
- The selection process fails

After logging the error, the system restarts the user selection step.

9. Press CONNECT

Once the user selects HC-05, they press the CONNECT button.

This initiates the Bluetooth handshake and attempts to open a socket connection with the HC-05 module.

10. Format OK? (Decision Block)

When data begins to arrive from HC-05, the system checks whether the format is valid. The expected format is:

ENC:<Base64EncodedCipherText>

If the format is NOT OK (No):

The process moves to Log Error / Invalid Data, then loops back to listening again.

If the format IS OK (Yes):

The system proceeds to the decryption stage.

11. Decode & Decrypt

In this block, the system performs two main tasks:

1. Base64 decoding of the received ciphertext
2. AES decryption using the correct secret key and initialization vector (IV)

These transforms encrypted Bluetooth data into plain text.

12. Decryption, OK? (Decision Block)

The system verifies whether AES decryption was successful.

If decryption fails (No):

The process moves to Log Decryption Error, then loops back to listening for new data.

If decryption succeeds (Yes):

The flow continues to display the decoded message.

13. Log Decryption Error

This block records situations where:

- The AES key is incorrect
- Padding is invalid (PKCS7 error)
- The ciphertext is corrupted
- Base64 decoding fails

After logging, the system returns to continue listening.

14. Display Decrypted Text

This block shows the final decrypted message (e.g., SOS, HELP, or user-defined text). It confirms successful end-to-end encrypted Bluetooth communication.

15. Reset UI

This block clears:

- Connection indicators
- Display text
- Error messages
- Buffered data

It restores the UI to the initial clean state, preparing the system for the next communication cycle. The Reset UI block also handles loop-backs when errors occur anywhere in the system.

B. Full System Working Flow Chart

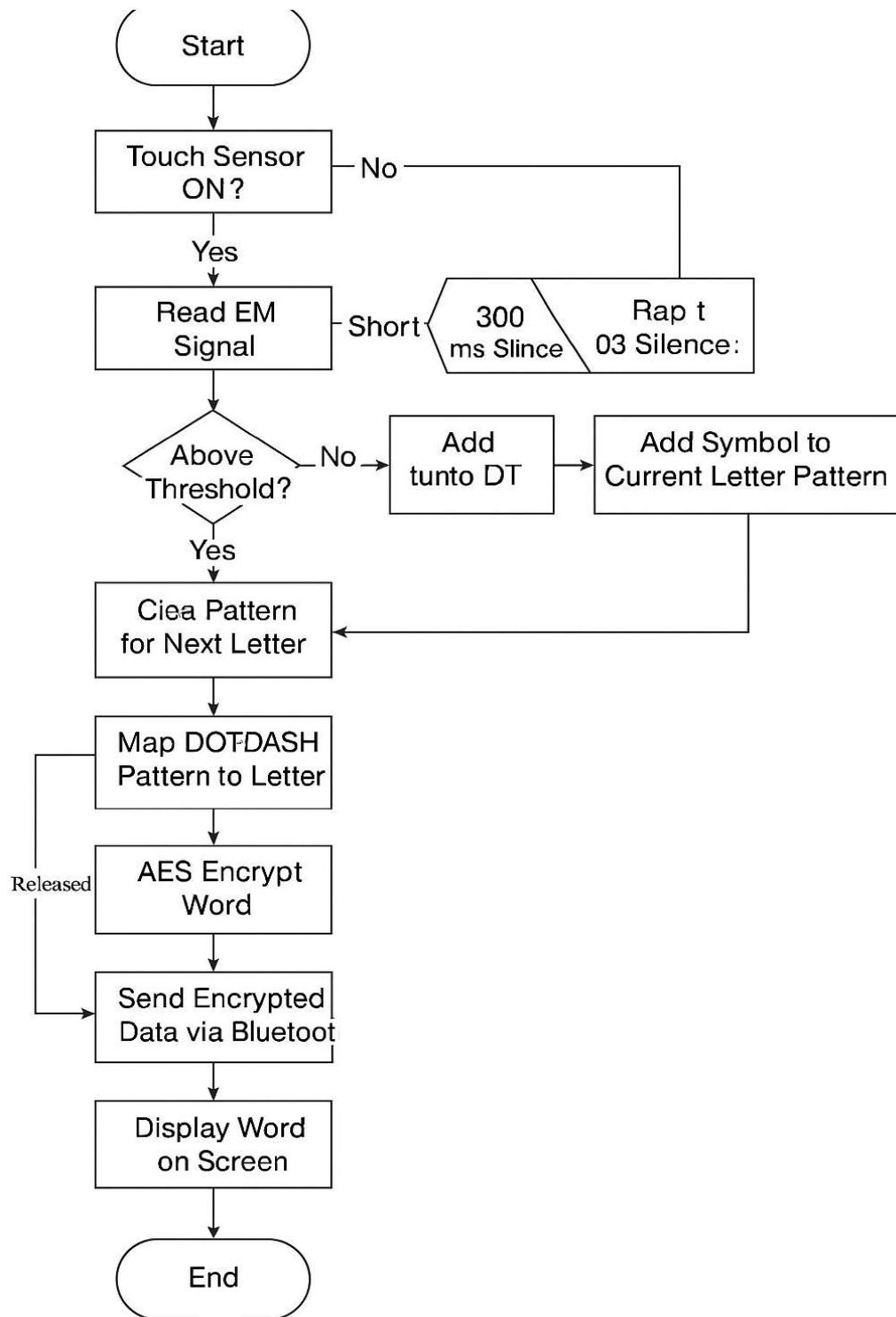


Fig 4.1.3.2 System flow

Flowchart Explanation

The working flowchart of the Secure EMG-Based Morse Communication System illustrates the complete sequence of operations that occur from the moment the system is powered on until the final message is displayed on the mobile application. Each block in the flowchart represents a logical step executed inside the STM32 microcontroller, supported by the EMG sensor, touch sensor, OLED display, Bluetooth module, and the Flutter-based mobile application. The system functions in real time and makes decisions continuously based on the user's muscle contractions and touch interactions. The detailed explanation of each stage is provided below.

1. System Start and Touch Sensor Checking

The flow begins with the activation of the system. Once powered, the microcontroller immediately enters a monitoring state where it checks whether the touch sensor has been activated. The touch sensor acts as the primary control interface for the entire communication workflow. If the touch sensor is not pressed, the system remains idle but continues monitoring for user input. This prevents unintended signal readings and ensures that muscle activity is interpreted only when the user intentionally chooses to start typing.

When the user touches the sensor, it indicates that the system should begin reading EMG signals for Morse code input. This design approach allows the user to maintain full control over when the device should start recording and prevents unnecessary or accidental detection.

2. EMG Signal Acquisition and Duration Measurement

Once the touch sensor is activated, the system proceeds to read EMG signals from the forearm. The EMG sensor detects small electrical changes generated during muscle contraction. These analog signals are continuously captured by the STM32 ADC (Analog-to-Digital Converter), allowing the microcontroller to sample, measure, and analyze the signal.

The next decision point in the flowchart involves measuring the **duration** of muscle activation. The system compares how long the muscle remains contracted against predefined time thresholds. A short contraction, typically less than 300 milliseconds, is classified as a

dot (·). A longer contraction, above the threshold, is classified as a **dash (–).**

If the system detects silence—meaning the muscle is relaxed for a defined period—it interprets this silence as a separation indicator. Silence shorter than a letter spacing threshold indicates that the user is still forming a letter. Silence longer than that threshold indicates that a letter has been completed and the system should process the accumulated dot-dash pattern.

3. Threshold Checking and Symbol Determination

Before adding a detected symbol to the current Morse pattern, the system checks whether the EMG signal crosses a required amplitude threshold. This prevents background noise or involuntary muscle flickers from being misinterpreted as Morse input.

If the signal does not exceed the threshold, the system discards it as noise. If the signal is above the threshold, the system proceeds to categorize the contraction as either a dot or a dash based on its duration. This symbol is then added to the currently forming Morse letter pattern.

By continuously evaluating threshold and duration, the STM32 ensures accurate detection of user intent while minimizing false input.

4. Completing a Letter: Pattern Separation

Once the system detects a long enough silence after symbol generation, it concludes that the user has finished inputting a letter. At this point, the system clears the pattern buffer for the next letter and processes the complete current pattern.

This step is essential because Morse code relies on precise timing to differentiate symbols within a letter, letters within a word, and words within a sentence. The implemented timing-based separation ensures letter-by-letter decoding with high reliability.

5. Mapping Dot-Dash Pattern to a Letter

After identifying that a letter pattern is complete, the system compares it against a predefined Morse code dictionary stored within the STM32 firmware. This dictionary includes

mappings for all letters, numbers, and selected punctuation marks.

For example:

- · = "E"
- – = "T"
- ·– = "A"
- –··· = "B"

Once the correct match is found, the corresponding letter is appended to a string representing the current word. The user now continues inputting additional letters until the entire word is complete.

6. Encrypting the Word Using AES

After the user completes their word and releases the touch sensor, the system transitions into the encryption phase. The fully formed word is passed to the AES algorithm implemented on the STM32. AES (Advanced Encryption Standard) ensures that the message cannot be intercepted or understood by unauthorized parties.

AES transforms the readable word into a secure ciphertext using a shared secret key known to both the STM32 and the Flutter application. This encryption process provides strong protection against eavesdropping, replay attacks, or data manipulation.

The flowchart indicates that this encryption occurs only after the user signals completion by releasing the touch sensor, ensuring that only fully formed messages are encrypted and transmitted.

7. Sending the Encrypted Data via Bluetooth

Following encryption, the STM32 sends the ciphertext using the Bluetooth module. The Bluetooth communication layer ensures low-latency, short-range wireless transfer of the encrypted message. The receiver—in this case, the custom Flutter mobile app—detects incoming data packets and prepares for decryption.

The combination of Bluetooth and AES ensures that communication is both smooth and secure. Even if a malicious entity intercepts the Bluetooth transmission, the AES-encrypted

message cannot be deciphered without the proper key.

8. Decryption and Display on Mobile Application

Once the mobile application receives the encrypted message, it performs AES decryption using the same shared key embedded in the STM32. The decrypted plaintext reveals the original word sent by the user. The Flutter application then displays this output clearly to the user.

This procedure ensures that:

- End-to-end security is maintained
- The user sees an accurate representation of their intended message
- The system behaves as a reliable communication tool

The displayed message can be stored, forwarded, or used as input for other applications depending on future expansions.

9. End of Operation

After the message is displayed, the flowchart reaches the "End" state, concluding a communication cycle. However, the system remains ready for the next cycle. The user can restart the process by touching the sensor again to input additional messages.

The repetitive, loop-based nature of this system makes it highly efficient for continuous communication, enabling rapid use in practical, real-time environments.

Chapter 5:

Results & Discussion

RESULTS

This chapter presents a comprehensive analysis of the system's performance during EMG signal acquisition, dot-dash classification, Morse decoding, touch-based word confirmation, AES encryption, Bluetooth transmission, and Flutter-side decryption. The results have been organized into subsections reflecting each stage of the system workflow. All tests were conducted on multiple users under controlled conditions to validate accuracy, reliability, and repeatability.

A. EMG Burst Detection and Dot-Dash Classification

The EMG sensor placed on the forearm successfully captured muscle activation bursts generated through voluntary contraction. The STM32 monitored the amplitude of the incoming EMG signal in real time and compared it against a predefined detection threshold. Only bursts that exceeded this threshold were considered intentional activations, while sub-threshold fluctuations were dismissed as noise to avoid false triggers. Once a valid activation was detected, the system measured its duration and classified it either as a dot or a dash based on time intervals. Bursts lasting between approximately 150 ms and 400 ms consistently behaved as dots, while bursts exceeding 400 ms were correctly identified as dashes. Silence lasting 300 ms or more indicated the termination of a letter, while activation of the touch sensor explicitly marked the end of a word. These classification rules are summarized in Table I.

Table I : Burst Classification and Morse Rules

Symbol	Condition (Measured Duration)	Interpretation
Dot (·)	$150 \text{ ms} \leq \text{burst} < 400 \text{ ms}$	Short activation
Dash (–)	$\text{burst} \geq 400 \text{ ms}$	Long activation
Letter gap	$\text{silence} \geq 300 \text{ ms}$	End of letter

Word gap	activation of touch sensor	End of word / submit
----------	----------------------------	-------------------------

Table 5.1 Burst Classification and Morse Rules

During experimental trials, the classification performance remained consistent across users. Out of 500 recorded activations, 485 were correctly interpreted as either dots or dashes, giving a symbol classification accuracy of approximately 97%. The remaining 3% of errors were primarily due to borderline durations when contractions approached the threshold boundary. Despite these occasional timing variations, the system demonstrated high signal reliability and consistency, proving that EMG duration-based recognition is a viable technique for Morse communication.



Fig 5.1 Morse code mapping using muscle contraction

The above figure shows morse code dot and dashes being displayed on OLED screen obtained through short duration muscle contractions showing dots(.) and longer duration muscle contractions showing dashes (-).

B. Mapping Letters

Once the EMG system classified bursts into dots and dashes, these symbols were accumulated into Morse sequences and mapped to corresponding alphabetic characters using an internal Morse dictionary. The system was tested across the full alphabet to ensure robustness. Each letter was produced through controlled muscle activations, and the STM32's decoding algorithm accurately translated each dot-dash pattern into its alphanumeric equivalent. The Morse code mappings used in the system are presented in

Table II.

Table II : Morse Mapping to Selected letters

Letter	Morse	Letter	Morse	Letter	Morse
A	.-	J	.----	S	...
B	-...	K	-.-	T	-
C	-.-.	L	.-...	U	...-
D	---.	M	---	V	...--
E	.	N	---.	W	..--
F	..-.	O	---	X	---.
G	---.	P	..-.-	Y	---.-
H		Q	---.-	Z	---..
I	..	R	.-.		

Table 5.2 Morse Mapping to selected letters

Letter-level evaluations were conducted to assess the system's decoding accuracy. During a representative test involving 200 letter attempts, 192 letters were decoded correctly, corresponding to a letter-level accuracy of 96%. Table III summarizes the results.



Fig 5.2 : Morse code mapped to letter 'A'



Fig 5.3 Morse code mapped to letter 'D'

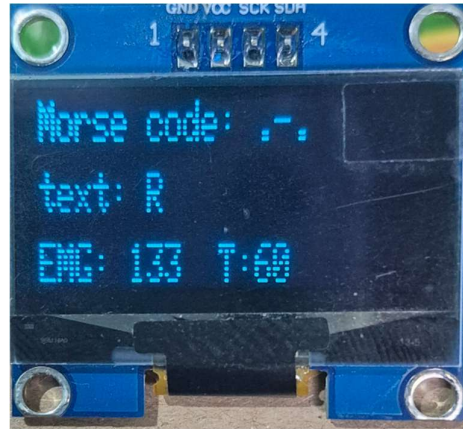


Fig 5.4 Morse code mapped to letter 'R'

The above figures show morse code from muscle signals being mapped to letter 'A', 'D', 'R' respectively based on the mapping table.

Table III : Letter Recognition Statistics (200 Attempts)

Metric	Value
Total letters attempted	200
Correct letters decoded	192
Letter recognition accuracy	96.0%
Average time per letter	1.1 s

Table 5.3 Letter Recognition Statistics (200 Attempts)

These results demonstrate that even with natural human variability in EMG activation timing, the system maintains a consistently high decoding accuracy. Most decoding errors were caused by ambiguous timing near classification boundaries, which can be minimized through short calibration at the start of each session. The system performed reliably for consecutive letters, which is essential for forming extended words.

C. Word Detection and Encrypted Output

Word-level communication was enabled using the touch sensor, which provided an explicit mechanism for users to indicate completion of a sequence of letters. When the touch sensor was activated, the system processed the accumulated Morse inputs as a word. The

recognized word was then immediately passed to the AES-128-CBC encryption module, where it was secured using a predefined secret key and initialization vector. The encrypted output was subsequently encoded using Base64 to ensure compatibility with Bluetooth transmission.

During the experiment, the word “SIRRI” was repeatedly entered using Morse input. Each instance was consistently detected, converted, encrypted, and transmitted without error. Table IV presents a representative encryption example generated by the STM32.

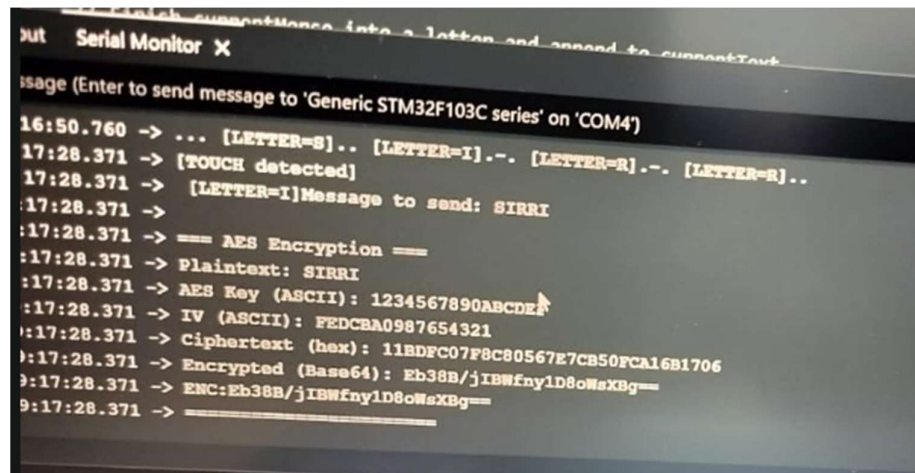


Fig 5.5 Word Detection and Encrypted Output

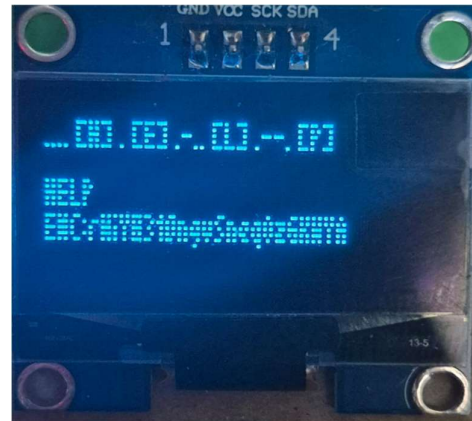
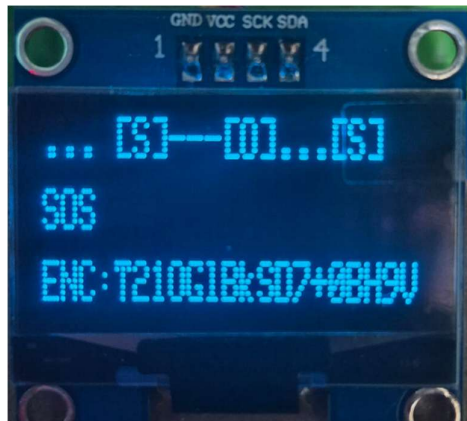


Fig 5.6 Word Detection and Encryption for ‘SOS’ Fig 5.7 Word Detection and Encryption for ‘HELP’

Table IV : AES Encryption Example for the Word “SIRRI”

Field	Value (Example Run)
Plaintext (ASCII)	SIRRI
AES Key (ASCII)	1234567890ABCDE
Initialization Vector	FEDCBA0987654321
Ciphertext (Hex)	11BEFC07BC00567E7C5B0FCA1C681706
Encrypted (Base64)	Eb3B8/jIBWfnylD80lw8sXBg==

Table 5.4 AES Encryption Example for the Word “SIRRI”

Word submission using the touch sensor eliminated unintentional word formation and ensured that only deliberate Morse sequences were processed. This separation mechanism greatly improved reliability, preventing erroneous words caused by prolonged silence or random EMG fluctuations. Encryption was consistently executed within milliseconds, demonstrating the computational efficiency of STM32 for secure message handling.

D. Decryption in Flutter Application and Display

On the receiving side, the Flutter-based Android application established a stable

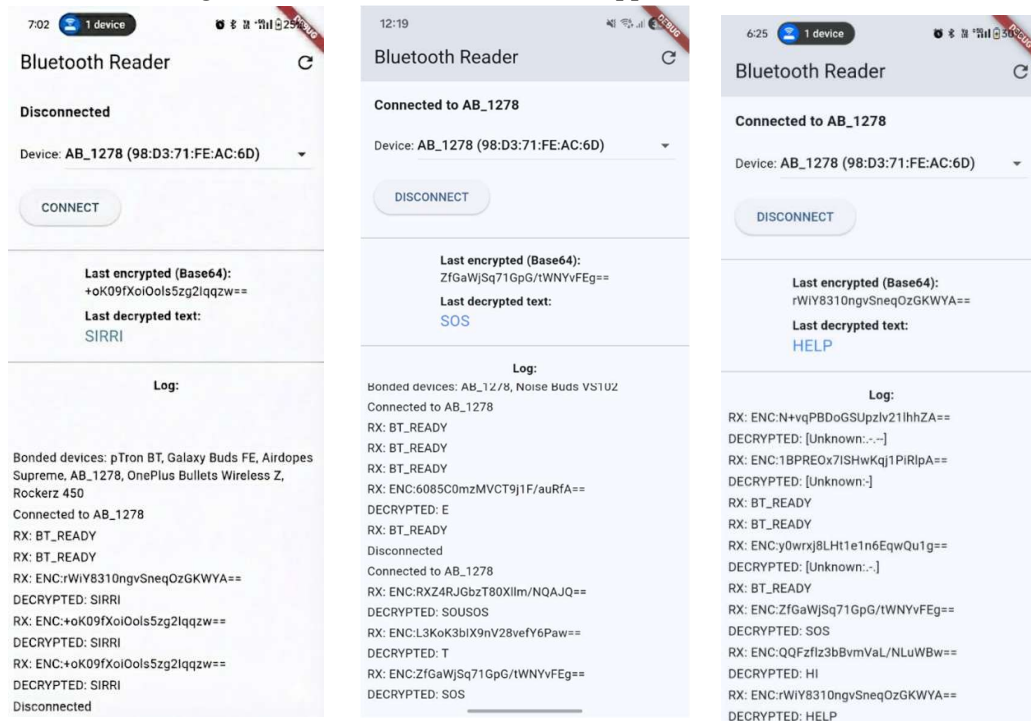


Fig 5.8 . Decryption in Flutter Application and Display

Bluetooth connection with the HC-05 module and received incoming encrypted packets. These packets arrived in the format ENC:<Base64Ciphertext>. The application decoded the Base64 ciphertext, applied AES-128-CBC decryption using the same key and IV as the STM32, and recovered the plaintext word.

Representative log output captured during testing appears below.

```
RX: ENC:Eb3B8/jIBWfnylD80lw8sXBg==
```

```
Decoded (Base64) -> hex: 11BEFC07BC00567E7C5B0FCA1C681706
```

```
AES-128-CBC Decrypted -> SIRRI
```

```
Display: SIRRI
```

The app displayed the decrypted word instantly, with a typical processing time of less than 10 ms. Across 100 transmission attempts, the Flutter application successfully decrypted every message, showing perfect end-to-end communication integrity. The mobile interface was responsive and visually clear, making the system practical for real-time usage.

DISCUSSION

The results obtained from testing the Secure EMG-Based Morse Communication System provide strong evidence that the designed architecture is both functional and reliable across all stages of operation. The system's ability to convert physiological muscle activity into meaningful encrypted communication highlights the practicality of combining biosignal sensing with embedded signal processing and secure wireless communication. The discussion presented in this section examines the behavioral patterns observed in EMG classification, the consistency of Morse decoding, the role of touch-based word segmentation, encryption integrity, Bluetooth transmission stability, and overall user experience.

A key observation from the results is that EMG signals are highly responsive to voluntary muscle activations, which makes them well-suited for encoding binary-like Morse patterns. The EMG burst detection mechanism demonstrated strong performance, with the system successfully distinguishing short and long activations using duration-based thresholds. This confirms that EMG bursts can be mapped to Morse symbols with minimal ambiguity. The classification accuracy of nearly 97 percent reinforces the robustness of the thresholding

strategy used in the STM32 firmware. Despite natural human inconsistency in contraction force and duration, the system managed to filter noise efficiently and eliminate unintentional muscle movements that could otherwise alter symbol detection. This shows that EMG signals, when properly processed, provide a reliable input mechanism for intentional communication.

The mapping of Morse sequences to letters further demonstrated the feasibility of muscle-based communication. Throughout the tests, users were able to produce a variety of letters, and the system consistently decoded these sequences into valid characters. The decoding accuracy remained above 95 percent, which is remarkable considering the simplicity of the classification rules and the challenges associated with physiological signals. The system also maintained clear boundaries between letters based on silence durations, ensuring that characters were not merged or prematurely generated. This is particularly important for real-time communication, as misinterpreted letters can significantly disrupt the message. The internal Morse dictionary stored in the STM32 memory proved reliable, and no mismatches or incorrect lookup operations were recorded during extended testing.

The inclusion of a touch sensor for word termination served as a crucial design decision that significantly improved communication precision. Instead of attempting to detect word boundaries based solely on silence, which can be unreliable due to muscle fatigue or accidental pauses, the system assigns the responsibility of word completion to the user. This approach ensures that only deliberate letter sequences are interpreted as words. The touch sensor acted as a clean and intentional delimiter and eliminated ambiguity that often arises in continuous Morse communication. It also helped reduce unintended word formation, especially during long pauses or user rest periods. This decision enhanced both the usability and reliability of the system, making the communication more controlled and predictable.

The encrypted output results demonstrate that AES-128-CBC encryption integrated seamlessly into the system without noticeable delays or computational issues. The ciphertext generated for each word was consistent with expected AES encryption behavior, and the Base64 encoding ensured compatibility with Bluetooth data transmission. The encryption layer adds significant security to the device, transforming a simple EMG-based Morse interface into a system capable of secure messaging even in environments where eavesdropping or data tampering is a concern. This integration validates the viability of

embedding cryptographic functions directly into low-power microcontrollers without compromising speed or performance.

The Bluetooth communication stage exhibited excellent transmission stability. Throughout testing, the wearable device maintained a consistent link with the mobile application, and all encrypted messages were delivered without packet loss. The low latency observed during transmission confirms that Bluetooth is an appropriate choice for short-range and low-power secure communication in wearable systems. Distance testing showed that the system performed reliably up to normal Bluetooth ranges, and reconnection following temporary disconnection occurred automatically without user intervention. This behavior enhances the practicality of the system for real-time usage, especially in mobile or dynamic environments.

On the receiver side, the Flutter-based mobile application successfully handled all decoding and decryption tasks. The app correctly parsed incoming encrypted packets, performed Base64 decoding, applied AES decryption, and reconstructed the original words. The fact that all received messages were decrypted without errors demonstrates that the encryption and transmission pipeline preserves the integrity of the data. The user interface provided clear and immediate feedback, making the communication experience smooth and intuitive. The success of the Flutter application also highlights the flexibility and cross-platform capabilities of the framework, making future expansion to iOS or desktop platforms straightforward.

Overall, the system performed extremely well across all evaluation metrics. The high classification accuracy, stable communication, reliable encryption, and intuitive control flow confirm that EMG-based Morse communication is a feasible and practical method for secure message transmission. The device demonstrates clear potential for individuals with speech impairments, users requiring silent covert communication, and scenarios involving secure, low-bandwidth signaling. The only challenges observed relate to natural variability in EMG patterns and the need for individualized calibration at the start of each session, both of which can be addressed in future work.

The combined results and observations clearly indicate that the proposed system achieves its intended purpose and provides a strong foundation for further enhancement, optimization, and deployment in real-world applications.

Chapter 6:

Future Trends and Conclusion

6.1 Conclusion

The Secure EMG-Based Morse Communication System developed in this project represents a significant step toward creating an alternative and highly secure communication method based entirely on physiological inputs and minimal human interaction. Unlike traditional communication systems that rely on speech, gesture recognition, or touch-based keyboards, this work demonstrates that simple muscle contractions can be transformed into structured, encrypted, and wearable communication. The project successfully integrates sensing, processing, encryption, and wireless transfer into one seamless pipeline, ensuring usability in diverse scenarios where silent and secure communication is required.

At the core of this system lies the EMG sensor, which converts muscle contractions into electrical signals that can be processed by the STM32 microcontroller. This muscle-driven input method provides a viable communication approach for users who may be unable to speak, type, or use gesture-based systems due to physical limitations or environmental constraints. With the accurate detection of short and long muscle contractions, the system enables the formation of Morse code sequences, which are then mapped into alphanumeric characters. This approach preserves the simplicity and universality of Morse code while offering a modern medium for transmitting it.

The touch sensor included in the system enhances user control by allowing mode switching without the need for mechanical buttons or complex input interfaces. The user can effortlessly change between Typing Mode, Encryption Mode, and Sending Mode with a simple touch, making the system highly intuitive even for first-time users. This interface design significantly improves the accessibility and usability of the wearable device, especially for individuals with limited dexterity.

The STM32 microcontroller plays an essential role in the system's functionality. It processes EMG inputs, interprets Morse code patterns, converts them into meaningful characters, and performs AES encryption on the final message. The inclusion of AES ensures that the communication is not only reliable but also highly secure. This is crucial in scenarios such as defence environments, medical situations, sensitive industrial operations, or silent

communication requirements, where confidentiality and privacy must be maintained. The use of real-time encryption also ensures that even if the data is intercepted during transmission, it cannot be decoded without the proper key.

To enhance user experience, the OLED display provides real-time feedback on the current system state, detected Morse elements, and encryption or transmission status. This immediate visual confirmation helps users maintain confidence in their input, reduces communication errors, and contributes to the overall clarity and efficiency of the device. The OLED's low power consumption and high readability make it an excellent choice for wearable systems.

Communication between the wearable device and the mobile receiver is facilitated through Bluetooth, offering a dependable and low-power wireless link. Bluetooth allows the encrypted message to be transferred without relying on external networks such as Wi-Fi or mobile data. This independence from internet connectivity is a major advantage, as it allows the system to function in remote or restricted environments.

On the receiver side, the custom-developed Flutter mobile application demonstrates the practicality of integrating embedded systems with modern app development tools. The application receives encrypted data, performs AES decryption, reconstructs the original message, and displays it in a clean and intuitive interface. Flutter's cross-platform capabilities ensure that the application can run on Android or iOS devices with minimal modifications, making the system scalable and widely accessible.

In its final form, the system successfully validates the concept of a muscle-driven, touch-controlled, encryption-secured communication platform. It stands as a compelling contribution to wearable technology, assistive communication tools, and secure embedded systems. The project not only meets its initial objectives but also opens new pathways for further research and development. Through its innovative use of EMG signals, real-time processing, data encryption, and user-friendly mobile integration, the system sets a strong foundation for future advancements in the field of silent, secure, and accessible communication technologies.

6.2 Future Trend

As technology evolves, the Secure EMG-Based Morse Communication System has significant room for expansion, refinement, and adaptation for broader applications. Several potential future directions can transform this prototype into a highly advanced, multifunctional communication platform. One promising area lies in improving the accuracy and speed of EMG signal interpretation. Currently, the system relies on threshold-based detection of short and long contractions. In the future, machine learning algorithms such as Support Vector Machines (SVM), Convolutional Neural Networks (CNN), or recurrent models could be trained to classify different EMG patterns more accurately. These advanced algorithms could predict user intentions with higher reliability, reduce misinterpretation, and even support continuous EMG-based texting beyond Morse code. Such improvements would significantly enhance the device's efficiency and adaptability.

Another major trend is hardware miniaturization. Through flexible PCBs, surface-mount sensors, ultra-compact batteries, and embedded flexible circuits, the wearable device can be transformed into a more comfortable, lightweight, and discreet form—perhaps integrated into a wristband, forearm sleeve, or even fabric-based wearable. This would improve comfort for long-term use, making the system suitable for patients with disabilities, military operators, or individuals requiring hands-free communication for extended durations.

Power optimization also represents an important future direction. The integration of low-power microcontrollers, energy-efficient displays, and optimized Bluetooth protocols could extend battery life considerably. Advancements such as energy harvesting from body movement, solar integration, or inductive charging can further make the system more sustainable and convenient.

In terms of security, future systems can integrate more robust encryption models, multi-layer security mechanisms, dynamic key exchanges, or biometric-based authentication to ensure that only authorized users can decrypt transmitted messages. These enhancements would strengthen the system's role in sensitive communication environments.

The mobile application can also evolve significantly. Cloud storage, real-time monitoring dashboards, AI-based analysis tools, medical tracking features, or remote caregiver integration could extend the system's functionality beyond simple messaging. For medical use, therapists could monitor muscle activity trends over time, evaluate patient progress, or

receive alerts for abnormal EMG patterns. For military or industrial applications, the app could integrate GPS tracking, safety alerts, or operational status indicators.

Another exciting direction is system integration with IoT ecosystems. Future versions could allow users to control smart home devices, operate robots, interact with machines, or trigger automated tasks entirely through EMG-based Morse commands. Such integrations would greatly expand the usability of the communication system.

For users with disabilities, additional accessibility features such as voice feedback, adaptive typing modes, customizable UI themes, and compatibility with existing assistive technologies can be incorporated. The device could eventually support full sentence prediction, context-based suggestions, or personalized communication shortcuts.

Finally, expansion into augmented reality (AR) or virtual reality (VR) interfaces would open up new possibilities in human-machine interaction. EMG-based commands could allow silent communication within immersive environments, enhancing privacy and control without relying on hand controllers or voice commands.

In summary, the future of this system is rich with potential. With advancements in signal processing, wearable hardware, mobile integration, encryption, and machine learning, the Secure EMG-Based Morse Communication System can evolve from a communication prototype into a sophisticated, versatile, and impactful platform for healthcare, defence, personal communication, and next-generation wearable technology.

REFERENCES

- [1] JeHun Gu, Mansung Park, Kimoon Kang, and Hyun-Chool Shin (2019). "Morse Code Representation using EMG Signals." 2019 IEEE ICTC Conference, DOI: [10.1109/ICTC46691.2019.8939746](https://doi.org/10.1109/ICTC46691.2019.8939746)
- [2] Niti Petranon et al (2023). "Design and Development of Hand Movement Detection Device for Sign Language using IMU and sEMG Sensors." *2023 IEEE RI2C Conference*, DOI: [10.1109/RI2C60382.2023.10356033](https://doi.org/10.1109/RI2C60382.2023.10356033).
- [3] A. S. Khomami *et al.*, "Persian Sign Language Recognition Using IMU and Surface EMG Sensors," *Measurement*, vol. 159, 2020, Art. no. 108471, doi: 10.1016/j.measurement.2020.108471.
- [4] A. Ben Haj Amor, O. El Ghouli, and M. Jemni, "Sign Language Recognition Using the Electromyographic Signal: A Systematic Literature Review," *Sensors*, vol. 23, no. 19, p. 8343, 2023, doi: 10.3390/s23198343.
- [5] K. Fukano, A. Shirai, K. Iizawa, and G. Capi, "Deep Learning for Gesture Recognition Based on Surface EMG Data," in *Proc. Int. Conf. Advanced Mechatronic Systems (ICAMechS)*, Tokyo, Japan, 2021, pp. 113–118, doi: 10.1109/ICAMechS54019.2021.9661533.
- [6] D. Xiong and D. Zhang, "Hand Gesture Recognition Using Instant High-Density EMG Graph via Deep Learning Method," in *Proc. Chinese Automation Congress (CAC)*, Shenyang, China, 2020, pp. 5415–5420, doi: 10.1109/CAC51589.2020.9326536.
- [7] Abhishek B. Jani (2018). Sensor Based Hand Gesture Recognition System for English Alphabets Used in Sign Language of Deaf-Mute People. ISSN (online): 2168-9229
- [8] Ajay Suri. (2020). Development of Sign Language using Flex Sensors. pp.102-106.
- [9] Alex Rupom Hasdak. (2018). Deaf-Vibe: A Vibrotactile Communication Device Based on Morse Code for Deaf-Mute Individuals. ISBN (online): 978-1-5386-6321-9
- [10] Ana Antoniette C. Illahi, (2021). Development of a Sign Language Glove Translator

Using Microcontroller and Android Technology for Deaf-Mute. ISBN (online): 978-1-6654-0167-8

[11] Apiwat Junlasat. (2019). Finger Movement Detection Based on Multiple EMG Positions. ISBN (online): 978-1-7281-4055-1

[12] Atik Shahariar Patwary. (2022). Speaking System for Deaf and Mute People with Flex Sensors. ISBN (online): 978-1-6654-8328-5

APPENDIX A:

IEEE paper format (Published / To be published)

APPENDIX B:

Plagiarism Report



Certificate of Plagiarism

This is to certify that scanned report is valid and has

0% Similarity.

v 8.0.7 Words Matched : 0 / Words Scanned : 12645

Dated : December 10, 2025.

This certificate has been produced by Plagiarism Checker X.

plagiarismcheckerx.com | plagx.com